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United States Patent Application Publication	20250282047
Kind Code	A1
Publication Date	September 11, 2025
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ROBOT

Abstract

A robot is disclosed. The robot includes: arms having at least one joint; an end effector coupled to one end of the arms; and a compensation coupler coupled to the end effector.

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Appl. No.:	18/860070
Filed (or PCT Filed):	April 27, 2022
PCT No.:	PCT/KR2022/006047

Publication Classification

Int. Cl.:	B25J9/10 (20060101)
U.S. Cl.:	
CPC	B25J9/104 (20130101);

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a robot. More specifically, the present disclosure relates to

a mechanism for payload compensation of a collaborative robot (co-robot or cobot).

BACKGROUND ART

[0002] A robot is a machine that can perform tasks autonomously. Robots have been an integral part to achieve production automation, and have been evolved into highly advanced intelligent robots. An intelligent robot refers to a robot that perceives the environment, makes decisions, and performs necessary tasks autonomously.

[0003] Intelligent robots are classified into two main categories: industrial robots and service robots. Unlike traditional industrial robots that typically work separately from humans, collaborative robots are designed to work alongside humans in a shared workspace by assisting or replacing human operators in performing tasks. Such collaborative robots are intended to physically interact with humans while performing tasks in a shared workspace, thereby improving the productivity. By contrast, service robots are designed to provide various kinds of services in social infrastructure, with the goal of enhancing the convenience of daily life.

[0004] Recently, numerous research has been conducted on intelligent robots.

DISCLOSURE

Technical Problem

[0005] It is an objective of the present disclosure to solve the above and other problems.

[0006] It is another objective of the present disclosure to provide a robot for performing various tasks.

[0007] It is yet another objective of the present disclosure to provide a robot for compensating for a payload in a task that involves the use of a heavy tool. It is yet another objective of the present disclosure to provide a payload compensation mechanism for heavy-duty tasks of a general-purpose collaborative robot.

Technical Solution

[0008] According to an aspect of the subject matter described in this application, a robot includes: arms having at least one joint; an end effector coupled to one end of the arms; and a compensation coupler coupled to the end effector.

ADVANTAGEOUS EFFECTS

[0009] A robot according to the present disclosure has the following effects.

[0010] According to at least one of the embodiments of the present disclosure, it is possible to provide a robot for performing various tasks.

[0011] According to at least one of the embodiments of the present disclosure, it is possible to provide a robot for compensating for a payload in performing a task that involves the use of a heavy tool.

[0012] According to at least one of the embodiments of the present disclosure, it is possible to provide a payload compensation mechanism for heavy-duty tasks of a general-purpose collaborative robot.

[0013] Further scope of applicability of the present disclosure will become apparent from the following detailed description. However, it should be understood that the detailed description and specific embodiments such as preferred embodiments of the present disclosure are given by way of example only, since various changes and modifications within the idea and scope of the present disclosure may be clearly understood by those skilled in the art.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0014] FIGS. **1** to **10** are diagrams illustrating examples of a robot and a payload compensation mechanism, according to embodiments of the present disclosure.

MODE FOR THE INVENTION

[0015] Description will now be given in detail according to exemplary embodiments disclosed herein, with reference to the accompanying drawings. For the sake of brief description with reference to the drawings, the same or equivalent components may be denoted by the same reference numbers, and description thereof will not be repeated.

[0016] In the following description, a suffix such as “module” and “unit” may be used to refer to elements or components. Use of such a suffix herein is merely intended to facilitate description of the specification, and the suffix itself is not intended to give any special meaning or function.

[0017] In the present disclosure, that which is well known to one of ordinary skill in the relevant art has generally been omitted for the sake of brevity. The accompanying drawings are used to help easily understand the technical idea of the present disclosure and it should be understood that the idea of the present disclosure is not limited by the accompanying drawings. The idea of the present disclosure should be construed to extend to any alterations, equivalents, and substitutes besides the accompanying drawings.

[0018] Although the terms “first,” “second,” etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another.

[0019] It will be understood that when a component is referred to as being “connected to” or “coupled to” another component, it may be directly connected to or coupled to another component, or intervening components may be present. In contrast, when a component is referred to as being “directly connected to” or “directly coupled to” another component, there are no intervening components present.

[0020] As used herein, a singular representation is intended to include a plural representation unless the context clearly indicates otherwise.

[0021] In the following description, when an embodiment is described with reference to a specific figure, reference numeral not shown in the specific figure may be mentioned, if necessary or desired. However, the reference numeral not shown in the specific figure may be mentioned only when the reference numeral is shown in other figures.

[0022] Referring to FIG. 1, a robot **10** may include a moving cart **20** and a robot arm **100**. The robot arm **100** may be mounted on the moving cart **20**. The robot arm **100** may be an articulated arm **100**. The moving cart **20** may provide coordinates to the robot arm **100**, and the robot arm **100** may move according to the coordinates provided by the moving cart **20**.

[0023] The moving cart **20** may include a cabinet **21** and a wheel **22**. The cabinet **21** may define an accommodation space therein. The robot arm **100** may be rotatably mounted on the cabinet **21**. The wheel **22** may be installed on a bottom of the cabinet **21**. The wheel **22** may allow the cabinet **21** to move freely.

[0024] A foot **23** may be installed on the bottom of the cabinet **21** while being adjacent to the wheel **22**. The foot **23** may limit movement of the cabinet **21**. The foot **23** may move in an up-and-down direction of the cabinet **21**. The foot **23** may be kept or stored inside the cabinet **21**, and its length may be increased as the foot **23** moves to an outside of the cabinet **21**. When the length of the foot **23** is increased, movement of the cabinet **21** may be inhibited, thereby allowing the cabinet **21** to be secured to a floor.

[0025] Referring to FIG. 2, the robot arm **100** may include a first arm **101**, a second arm **102**, and a third arm **103**. The first arm **101** may be referred to as a mount base **101**. The first arm **101** may be rotatably mounted on top of the moving cart **20** (see FIG. 1). The second arm **102** may be rotatably coupled to the first arm **101**. The third arm **103** may be rotatably coupled to the second arm **102**. A rotation axis of the third arm **103** may intersect a rotation axis of the second arm **102**.

[0026] The robot arm **100** may further include a fourth arm **104**. The fourth arm **104** may be rotatably coupled to the third arm **103**. A rotation axis of the fourth arm **104** may intersect the rotation axis of the third arm **103**. The rotation axis of the fourth arm **104** may be parallel to the rotation axis of the second arm **102**.

[0027] The robot arm **100** may further include a fifth arm **105**, a sixth arm **106**, a seventh arm **107**, an eighth arm **108**, and a ninth arm **109**. The number of arms included in the robot arm **100** is not limited thereto.

[0028] Referring to FIG. 3, an end effector **110** may be coupled to the robot arm **109**. A first tool **120** may be coupled to the end effector **110**. The first tool **120** may be fixed to the end effector **110** so as to move together with the end effector **110**. For example, when the end effector **110** rotates, the first tool **120** may rotate together with the end effector **110**.

[0029] A second tool **200** may be coupled to the first tool **120**. A third tool **300** may be coupled to the second tool **200**. The third tool **300** may rotate together with the first tool **120**. The second tool **200** may be referred to as a coupler **200** or a compensation coupler **200**. The third tool **300** may be referred to as a load base **300**.

[0030] When the end effector **110** rotates, the first tool **120** and the third tool **300** may rotate together with the end effector **110**. The third tool **300** and the first tool **120** may be connected to each other by the second tool **200**. The third tool **300** and the first tool **120** may move relative to the second tool **200**. The first tool **120** may be coupled or fixed to the end effector **110** by a fastening member **f2**.

[0031] Referring to FIGS. 4 to 6, the compensation coupler **200** may include a case **210**, a rotating shaft **230**, a bearing **240, 250**, and a cover **220**. The case **210** may be of a cylindrical shape, and may have openings at opposite ends thereof. The openings at the opposite ends of the case **210** may have different diameters. The case **210** may include a body **211** and an end rim **212**. The body **211** may be of a cylindrical shape, and the end rim **212** may be a donut-shaped disc (or disk). An outer diameter of the end rim **212** may be substantially equal to an outer diameter of the body **211**, and an inner diameter of the end rim **212** may be less than an inner diameter of the body **211**.

[0032] The bearing **240, 250** may include an inner ring **241, 251**, an outer ring **242, 252**, and a roller **243, 253**. A diameter of the outer ring **242, 252** may be greater than a diameter of the inner ring **241, 251**, and the roller **243, 253** may be inserted between the outer ring **242, 252** and the inner ring **241, 251**. The inner ring **241, 251** may rotate relative to the outer ring **242, 252**. The bearing **240, 250** may be inserted into the case **210**. The outer ring **242, 252** of the bearing **240, 250** may be in contact with or fixed to an inside of the body **211**. The outer ring **242, 252** of the bearing **240, 250** may be supported by the end rim **212** of the body **211**. The end rim **212** of the body **211** may be in contact with the outer ring **242, 252** of the bearing **240, 250**. There may be a plurality of bearings **240** and **250**.

[0033] The rotating shaft **230** may be inserted into the case **210** so as to rotate therein. The rotating shaft **230** may be inserted into the inner ring **241, 251** of the bearing **240, 250**. The rotating shaft **230** may be press-fitted into or fixed to the inner ring **241, 251** of the bearing **240, 250**. The rotating shaft **230** may include a shaft body **231** and a flange **232**. The shaft body **231** may be an elongated cylinder, and may be inserted into the inner ring **241, 251** of the bearing **240, 250**. The shaft body **231** may rotate together with the inner ring **241, 251** of the bearing **240, 250**. The flange **232** may be provided at or fixed to a first end of the shaft body **231**. The flange **232** may have a disk shape. An outer diameter of the flange **232** may be greater than an outer diameter of the shaft body **231**. The flange **232** may be supported by or in contact with the inner ring **241, 251** of the bearing **240, 250**. A groove **234** may be formed in an outer circumferential surface of the shaft body **231** while being adjacent to a second end of the shaft body **231**. A lock washer **201** may be inserted into the groove **234** of the shaft body **231**. When the lock washer **201** is fitted into the groove **234** of the shaft body **231**, the lock washer **201** may come into contact with the inner ring **241, 251** of the bearing **240, 250**. Accordingly, the rotating shaft **230** may be fixed to the inner ring **241, 251** of the bearing **240, 250**.

[0034] The cover **220** may be coupled to the body **211** of the case **210**. The cover **220** may fix the bearing **240, 250** inside the case **210**. The cover **220** may be disposed opposite the end rim **212** with respect to the bearing **240, 250**.

[0035] A coupling hole **233** may be formed in an outer circumferential surface of the flange **232**. The coupling hole **233** may be formed in a radial direction of the flange **232**. The coupling hole **233** may be provided in plurality. The first tool **120** may be coupled or fixed to the flange **232**. The first tool **120** may include a shaft **121**, a flange **122**, and a fastening hole **123**. The coupling hole **233** may extend to the shaft **121** of the first tool **120**. The fastening hole **233** may be formed in the flange **232** in an axial direction. A first fastening member **f1** (see FIG. 3) may be inserted into the coupling hole **233** to allow the rotating shaft **230** and the first tool **120** to be coupled to each other, and a second fastening member **f2** (see FIG. 3) may be inserted into the coupling hole **123** to allow the first tool **120** and the end effector **110** (see FIG. 3) to be coupled to each other.

[0036] A swivel **260** may be rotatably coupled to the case **210**. A plurality of swivels **260** may be provided. The plurality of swivels **260** may include a first swivel **260A** and a second swivel **260B**. The first swivel **260A** may be spaced apart from the second swivel **260B** in a circumferential direction of the case **210**. The description of the first swivel **260A** is equally applicable to the second swivel **260B**.

[0037] The swivel **260** may include a rotary body **261**, bolts **262** and **264**, and a biner **263**. The rotary body **261** may be rotatably coupled to an outer circumferential surface of the case **210** by the bolts **262** and **264**. The rotary body **261** may rotate about the bolts **262** and **264**. The biner **263** may be pivotably coupled to the rotary body **261**. For example, the biner **263** may be U-shaped. Both ends of the biner **263** may be pivotably coupled to the rotary body **261** by the bolts **262** and **264**.

[0038] Referring to FIG. 7, the first tool **120** may be coupled to the end effector **110** of the arms **106**, **107**, **108** and **109**, the compensation coupler **200** may be coupled to the first tool **120**, and the third tool **300** may be coupled to the compensation coupler **200**. A cable **CA** may be hooked to the swivel **260A**, **260B** by a cable hook **CH**. A first cable **CA** may be coupled to the first swivel **260A**, and a second cable **CA** may be coupled to the second swivel **260B**. The cables **CA** may pull the compensation coupler **200**. In this case, the force with which the cables **CA** pull the compensation coupler **200** may correspond to the weight of the third tool **300**. Accordingly, the arms **106**, **107**, **108** and **109** and/or the end effector **110** may be operated regardless of the weight of the third tool **300**.

[0039] Referring to FIG. 8 along with FIG. 7, a rail **R1**, **R2** may be installed at a fixture (or structure). There may be a plurality of rails **R1** and **R2**. A first rail **R1** may be fixed to the fixture, and a second rail **R2** may be disposed in parallel with the first rail **R1** and fixed to the fixture. A moving block **S1**, **S2** may be movable on the rail **R1**, **R2**. A first moving block **S1** may be movable on the first rail **R1**, and a second moving block **S2** may be movable on the second rail **R2**. A first mount **M1** may be mounted on the first moving block **S1**. A second mount **M2** may be mounted on the second moving block **S2**. A strut bar **SB** may connect the first mount **M1** and the second mount **M2**.

[0040] A third swivel **H1** may be pivotably coupled to the first mount **M1**, and a fourth swivel **H2** may be pivotably coupled to the second mount **M2**. A first spring balancer **T1** may be coupled to the third swivel **H1**, and a second spring balancer **T2** may be coupled to the fourth swivel **H2**. The first cable **CA** may be wound on or unwound from the first spring balancer **T1**, and the second cable **CA** may be wound on or unwound from the second spring balancer **T2**.

[0041] Referring to FIGS. 9 and 10, the compensation coupler **200** may offset the weight of a load on the rotating shaft **230** by the cables **CA**. For example, the sum of forces (the vector sum of forces **F1**, **F2** and **F3**) acting on the rotating shaft **230** may be zero. The tension of the spring balancers **T1** and **T2** may correspond to the load acting on the rotating shaft **230**. Accordingly, the end effector **110** may be moved by the arms **100** regardless of the weight of the load on the rotating shaft **230**.

[0042] When the robot arm **100** is operated, coordinates of the end effector **110** may be variously defined. As the end effector **110** moves, the cable **CA** and the spring balancer **T1**, **T2**, which are coupled to the compensation coupler **200**, may move on the rail **R1**, **R2**, thereby achieving the

degree of freedom of movement of the robot arm **100**.

[0043] The robot arm **100** may be moved into a workspace where human operators work for collaboration. Regardless of the weight of a tool required for a task, the general-purpose robot arm **100** may replace humans in performing tasks or collaborate with humans to perform tasks.

[0044] Referring to FIGS. **1** to **10**, a robot may include: arms having at least one joint; an end effector coupled to one end of the arms; and a compensation coupler coupled to the end effector. The compensation coupler may include: a case that defines an accommodation space therein; a rotating shaft passing through the accommodation space and configured to be rotatable; and a bearing positioned between the case and the rotating shaft. The rotating shaft may be coupled to the end effector.

[0045] The compensation coupler may include a plurality of swivels rotatably coupled to the case. Each of the plurality of swivels may include: a rotary body configured to rotate on the case; and a biner pivotably coupled to the rotary body.

[0046] A pivot axis of the biner may intersect a rotation axis of the rotary body.

[0047] The case may include: a body having a cylindrical shape, with openings at opposite ends of the body; an end rim formed at the opening of a first end of the body, the end rim having an inner diameter less than an inner diameter of the body; and a cover coupled to the opening of a second end of the body, the cover having an inner diameter less than the inner diameter of the end rim. The bearing may include: an outer ring in contact with the body and the end rim; an inner ring disposed within the outer ring; and a roller inserted between the outer ring and the inner ring.

[0048] The rotating shaft may include: a shaft body press-fitted into the inner ring of the bearing; and a flange disposed adjacent to the end rim of the case and provided at a first end of the shaft body, the flange having an outer diameter greater than an outer diameter of the shaft body. The inner ring of the bearing may be in contact with the flange.

[0049] The compensation coupler may further include a lock washer that is disposed adjacent to the cover, fixed to an outer circumferential surface of the shaft body of the rotating shaft, and in contact with the inner ring of the bearing.

[0050] The robot may further include: a tool fixed to the rotating shaft; and a plurality of cables coupled to the plurality of swivels, respectively. A resultant force of the plurality of cables may correspond to a weight of the tool.

[0051] The robot may further include: a plurality of rails; and a plurality of moving blocks movable on the plurality of rails, respectively. The plurality of cables may be connected to the plurality of moving blocks, respectively.

[0052] A payload compensation system of a robot may include: a plurality of fixed rails; a plurality of moving blocks movable on the plurality of rails, respectively; a plurality of cables connected to the plurality of moving blocks, respectively; the robot having an end effector and including arms having at least one joint; and a compensation coupler coupled to the end effector and connected to the plurality of cables.

[0053] The plurality of cables may be respectively coupled to the plurality of moving blocks by a respective plurality of spring balancers.

[0054] The compensation coupler may include: a case that defines an accommodation space therein; a rotating shaft passing through the accommodation space and configured to be rotatable; and a bearing positioned between the case and the rotating shaft. The rotating shaft may be coupled to the end effector.

[0055] The compensation coupler may include a plurality of swivels rotatably coupled to the case. Each of the plurality of swivels may include: a rotary body configured to rotate on the case; and a biner pivotably coupled to the rotary body. The plurality of cables may be coupled to the plurality of biners, respectively.

[0056] A pivot axis of the biner may intersect a rotation axis of the rotary body.

[0057] The payload compensation system may further include a tool fixed to the rotating shaft. A

resultant force of the plurality of cables may correspond to a weight of the tool.

[0058] Certain embodiments or other embodiments of the disclosure described above are not mutually exclusive or distinct from each other. Any or all elements of the embodiments of the disclosure described above may be combined or combined with each other in configuration or function.

[0059] For example, a configuration “A” described in one embodiment of the disclosure and the drawings, and a configuration “B” described in another embodiment of the disclosure and the drawings may be combined with each other. Namely, although the combination between the configurations is not directly described, the combination is possible except in the case where it is described that the combination is impossible.

[0060] The above detailed description is to be construed in all aspects as illustrative and not restrictive. The scope of the present disclosure should be determined by reasonable interpretation of the appended claims, and all changes coming within the equivalency range of the present disclosure are intended to be embraced in the scope of the present disclosure.

Claims

1. A robot comprising: arms having at least one joint; an end effector coupled to one end of the arms; and a compensation coupler coupled to the end effector, wherein the compensation coupler comprises: a case that defines an accommodation space therein; a rotating shaft passing through the accommodation space and configured to be rotatable; and a bearing positioned between the case and the rotating shaft, and wherein the rotating shaft is coupled to the end effector.
2. The robot of claim 1, wherein the compensation coupler includes a plurality of swivels rotatably coupled to the case, and wherein each of the plurality of swivels comprises: a rotary body configured to rotate on the case; and a biner pivotably coupled to the rotary body.
3. The robot of claim 2, wherein a pivot axis of the biner intersects a rotation axis of the rotary body.
4. The robot of claim 2, wherein the case comprises: a body having a cylindrical shape, with openings at opposite ends of the body; an end rim formed at the opening of a first end of the body, the end rim having an inner diameter less than an inner diameter of the body; and a cover coupled to the opening of a second end of the body, the cover having an inner diameter less than the inner diameter of the end rim, and wherein the bearing comprises: an outer ring in contact with the body and the end rim; an inner ring disposed within the outer ring; and a roller inserted between the outer ring and the inner ring.
5. The robot of claim 4, wherein the rotating shaft comprises: a shaft body press-fitted into the inner ring of the bearing; and a flange disposed adjacent to the end rim of the case and provided at a first end of the shaft body, the flange having an outer diameter greater than an outer diameter of the shaft body, and wherein the inner ring of the bearing is in contact with the flange.
6. The robot of claim 5, further comprising a lock washer disposed adjacent to the cover and fixed to an outer circumferential surface of the shaft body of the rotating shaft, the lock washer being in contact with the inner ring of the bearing.
7. The robot of claim 2, further comprising: a tool fixed to the rotating shaft; and a plurality of cables coupled to the plurality of swivels, respectively, wherein a resultant force of the plurality of cables corresponds to a weight of the tool.
8. The robot of claim 7, further comprising: a plurality of rails; and a plurality of moving blocks movable on the plurality of rails, respectively, wherein the plurality of cables are connected to the plurality of moving blocks, respectively.
9. A payload compensation system of a robot, the payload compensation system comprising: a plurality of fixed rails; a plurality of moving blocks movable on the plurality of rails, respectively; a plurality of cables connected to the plurality of moving blocks, respectively; the robot having an

end effector and including arms having at least one joint; and a compensation coupler coupled to the end effector and connected to the plurality of cables.

10. The payload compensation system of claim 9, wherein the plurality of cables are respectively coupled to the plurality of moving blocks by a respective plurality of spring balancers.

11. The payload compensation system of claim 10, wherein the compensation coupler comprises: a case that defines an accommodation space therein; a rotating shaft passing through the accommodation space and configured to be rotatable; and a bearing positioned between the case and the rotating shaft, and wherein the rotating shaft is coupled to the end effector.

12. The payload compensation system of claim 11, wherein the compensation coupler includes a plurality of swivels rotatably coupled to the case, wherein each of the plurality of swivels comprises: a rotary body configured to rotate on the case; and a biner pivotably coupled to the rotary body, and wherein the plurality of cables are coupled to the plurality of biners, respectively,

13. The payload compensation system of claim 12, wherein a pivot axis of the biner intersects a rotation axis of the rotary body.

14. The payload compensation system of claim 11, further comprising a tool fixed to the rotating shaft, wherein a resultant force of the plurality of cables corresponds to a weight of the tool.
